

Supplementary Information

Stability certificates for selective reasoning over evolving knowledge graphs

Evaluation record

Public runs use 5,000 evenly sampled validation and test point queries, five validation permutations, a fixed ridge penalty of 0.01, half-life 35 timestamp units and bins of seven timestamp units. Neural results use three trained checkpoints. Chronological validation is split in time into calibration and threshold periods. All reported evidence diagnostics use the corrected full-mass deletion calculation. Original runs are preserved separately. The full

Table S1: Public results. B/S denotes baseline/StableKG. Coverage targets are set on validation data; realized test coverage can differ.

Dataset	MRR	Brier B/S	ECE B/S	AURC B/S	Risk20 B/S	Risk40 B/S
ICEWS14	0.3153	0.1560/0.1426	0.0405/0.0158	0.5872/0.5494	0.5079/0.4350	0.5630/0.5308
ICEWS05-15	0.4221	0.1889/0.1720	0.0410/0.0189	0.5011/0.4669	0.4039/0.3155	0.4850/0.4272
GDELT	0.2454	0.1134/0.1122	0.0159/0.0160	0.7694/0.7647	0.7077/0.7032	0.7587/0.7646

Table S2: Paired differences (StableKG minus baseline), seed 0, 1,000 resamples of contiguous timestamp blocks. Risk uses frozen validation thresholds separately for each method; these are not equal accepted-count comparisons.

Dataset	Metric	Estimate	2.5%	97.5%
ICEWS14	risk_delta@0.20	-0.0420	-0.0659	-0.0171
ICEWS14	risk_delta@0.40	-0.0352	-0.0523	-0.0192
ICEWS14	auroc_delta	-0.0365	-0.0446	-0.0287
ICEWS05-15	risk_delta@0.20	-0.0858	-0.1110	-0.0599
ICEWS05-15	risk_delta@0.40	-0.0514	-0.0704	-0.0308
ICEWS05-15	auroc_delta	-0.0353	-0.0438	-0.0274
GDELT	risk_delta@0.20	0.0020	-0.0184	0.0207
GDELT	risk_delta@0.40	0.0099	0.0017	0.0195
GDELT	auroc_delta	-0.0039	-0.0093	0.0013

feature model contains a redundant linear combination because $c = (f + d)/2$. Its inclusion affects ridge geometry, not the information available. Paired block-bootstrap contrasts for each control are included in the source CSV files. These comparisons are diagnostic; no model was selected using the test results.

Table S3: Matched-control AUC over five validation permutations. B contains margin, log-support and recency. All configurations share the same calibration/operating splits and ridge penalty. The entire feature grid was retained.

Features	ICEWS14	ICEWS05-15	GDELT
Margin only	0.5681	0.4960	0.7932
B	0.5872	0.5011	0.7694
B + c	0.5666	0.4818	0.7703
B + z	0.5503	0.4700	0.7657
B + f, d	0.5641	0.4777	0.7687
B + z, f	0.5504	0.4696	0.7657
B + z, d	0.5490	0.4682	0.7648
B + z, f, d	0.5487	0.4667	0.7648
Full	0.5494	0.4669	0.7647
B + k_*	0.5745	0.4722	0.7692
B + softmax, entropy	0.5809	0.4932	0.7702

Table S4: Intervention flip rates in low/high diagnostic-score quartiles. The window spans seven timestamp units. Queries with no winner evidence are excluded, and tied quartiles can contain different counts.

Dataset	Queries	Single event	Window	Counter-event
ICEWS14	3653	0.8414/0.0011	0.9980/0.0011	0.8924/0.1171
ICEWS05-15	4368	0.8566/0.0000	0.9978/0.0000	0.9158/0.1648
GDELT	4981	0.6100/0.0000	0.9997/0.0000	0.6615/0.0120

Table S5: Neural and chronological results. B/S denotes baseline/StableKG.

Protocol	Dataset	AUC B/S	Risk20 B/S	Risk40 B/S
Neural	ICEWS14	0.5954/0.5570	0.4897/0.4203	0.6006/0.5336
Neural	ICEWS05-15	0.5437/0.4701	0.4346/0.2841	0.5382/0.4376
Chronological	ICEWS14	0.4705/0.4411	0.3665/0.2686	0.4373/0.4203
Chronological	ICEWS05-15	0.4254/0.3957	0.3222/0.2342	0.4106/0.3618

Table S6: Recent author-code TeRDy comparison. Metrics are from the completed GPU runs under the shared Pe panel.

Dataset	Rank	Epochs	MRR	AURC B	AURC S	Risk20 B/S
ICEWS14	6000	12	0.6571	0.3801	0.2896	0.1880/0.1530
ICEWS05-15	2000	8	0.5355	0.3801	0.3716	0.2330/0.2210

Table S7: Public update maintenance for ICEWS14. Means over 10 repetitions; all outputs matched the full refresh.

Events	Decisions	Batch	Full ms	Incremental ms	Refreshed
72,827	21,667	100	136.3	1.3	98.2

Table S8: Public update maintenance for ICEWS05-15. Means over 10 repetitions; all outputs matched the full refresh.

Events	Decisions	Batch	Full ms	Incremental ms	Refreshed
368,963	51,292	100	363.6	2.2	97.7

Table S9: Public update maintenance for GDELT. Means over 10 repetitions; all outputs matched the full refresh.

Events	Decisions	Batch	Full ms	Incremental ms	Refreshed
2,308,166	9,741	100	223.0	4.5	98.7

Table S10: Controlled rule-DAG update study, 20 seeds. Runtime excludes closure discovery and measures propagation over a cached closure. Runtime and node counts use different changed-base samples.

Metric	Mean	2.5%	97.5%
Node reduction	71.56%	70.73%	72.33%
Propagation-time reduction	47.67%	42.22%	52.84%
Maximum output discrepancy	0.0	0	0

Table S11: Intervention-survival AUROC for matched diagnostics. Higher is better. c is the deletion diagnostic and z the normalized margin; raw margin provides the control for a fixed injected event.

Dataset	Edit	c	z	Margin	Support
ICEWS14	event	0.8767	0.6577	0.9373	0.9066
ICEWS14	window	0.9977	0.6919	0.9193	0.8947
ICEWS14	counter	0.8714	0.6480	0.9999	0.9092
ICEWS05-15	event	0.9053	0.6273	0.8742	0.8527
ICEWS05-15	window	0.9982	0.6747	0.8488	0.8421
ICEWS05-15	counter	0.8795	0.5495	1.0000	0.8720
GDELT	event	0.8033	0.8946	0.9930	0.8318
GDELT	window	0.9997	0.9034	0.9579	0.8313
GDELT	counter	0.8124	0.8800	1.0000	0.8434

Table S12: Exactly equal accepted-count comparison, calibration seed 0. Both frozen score orderings accept the same number of test queries. Coverage budgets (20%, 40%) are prespecified; ties retain fixed source order. Intervals use 1,000 paired timestamp-block resamples and recompute each ordering at that budget. This evaluates a fixed budget, separately from the deployable validation-threshold policy.

Dataset	Coverage	Risk B	Risk S	Difference	2.5%	97.5%
ICEWS14	20%	0.5060	0.4390	-0.0670	-0.1036	-0.0314
ICEWS14	40%	0.5650	0.5325	-0.0325	-0.0501	-0.0165
ICEWS05-15	20%	0.4070	0.3230	-0.0840	-0.1069	-0.0606
ICEWS05-15	40%	0.4880	0.4410	-0.0470	-0.0678	-0.0307
GDELT	20%	0.7070	0.7090	0.0020	-0.0193	0.0212
GDELT	40%	0.7575	0.7635	0.0060	-0.0025	0.0147

Reproduction and interpretation

Run `run_all.ps1 -RunSynthetic -RunNeural -CompilePaper` to regenerate the experiments, statistics, figures and manuscript. The local checkpoint refresh command is recorded in `scripts/refresh_evidence.py`. Tests cover causal cutoffs, filtered ranking, actual predict-before-reveal replay, the discrete window-count guarantee, floating-point full-mass deletion and incremental/full agreement.

The public interpolation-style split permits training events at the query timestamp. Chronological replay instead predicts before revealing that timestamp. Selective correctness means that the unfiltered top prediction belongs to the published query answer set; filtered MRR separately removes other known same-query answers. Thus selective correctness and filtered Hits@1 have different evaluation denominators. Timestamp units are dataset indices and are not assumed to be calendar days. The additive-scorer certificate does not certify neural-parameter robustness.

The prefix attack minimizes the number of deleted windows, not their mass over arbitrary subsets. For masses (6, 4, 3) and margin 7 it removes two windows with mass 10, whereas another two-window deletion has mass 7. This distinction fixes the interpretation of the guarantee without treating the continuous diagnostic as a universal robustness radius.

Source files

Corrected public outputs are in `results_final_v5/`; interventions, neural inference and chronological replay have separate version-5 directories. Matched controls and raw feature caches are in `results_review_20260909/`. Each table here is generated from those files. Bibliographic metadata responses and retrieval errors are retained under `literature/verification_20260909/`.